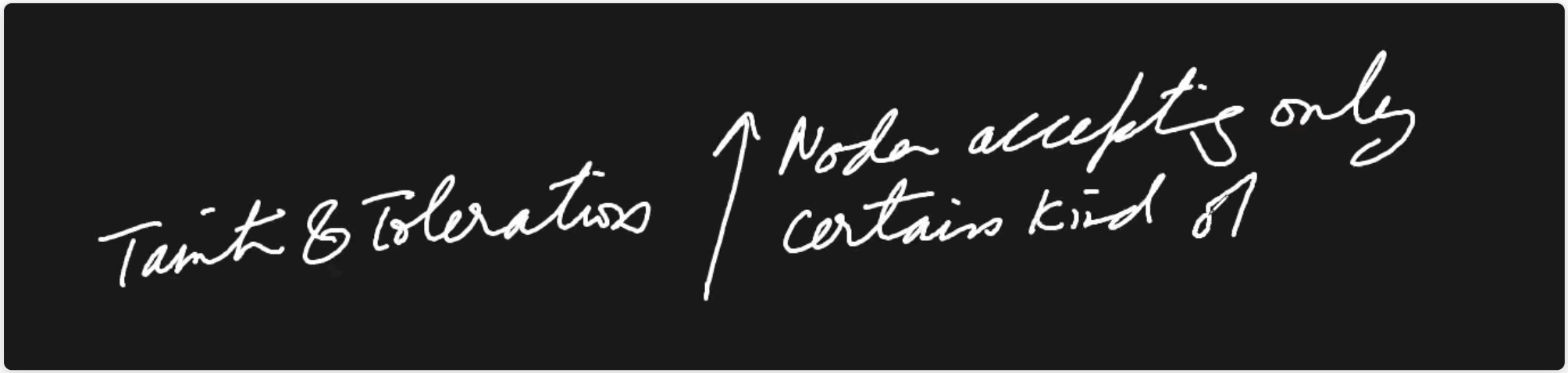
KUBERNETES CKA PREP › MY LEARNING NOTES › SESSION 07

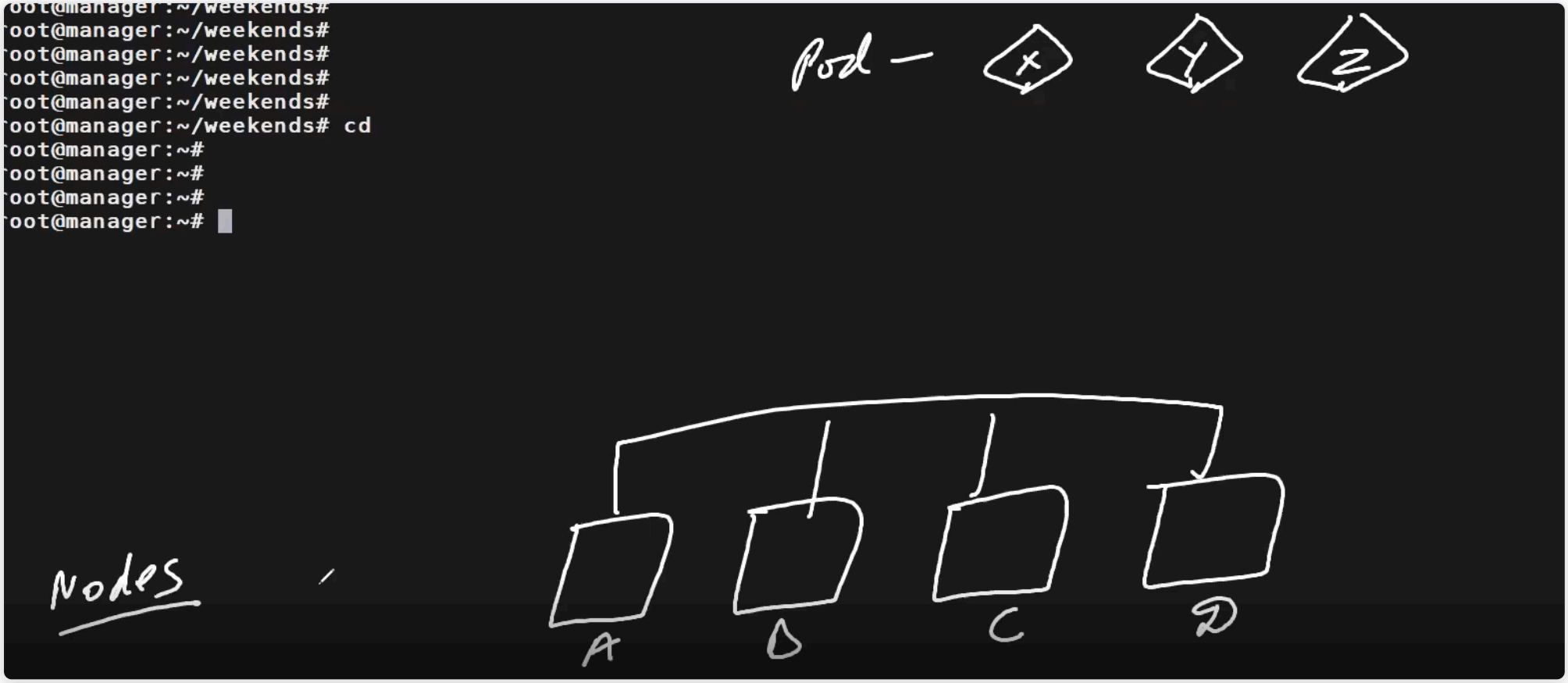
Screenshots: Taints and Tolerations

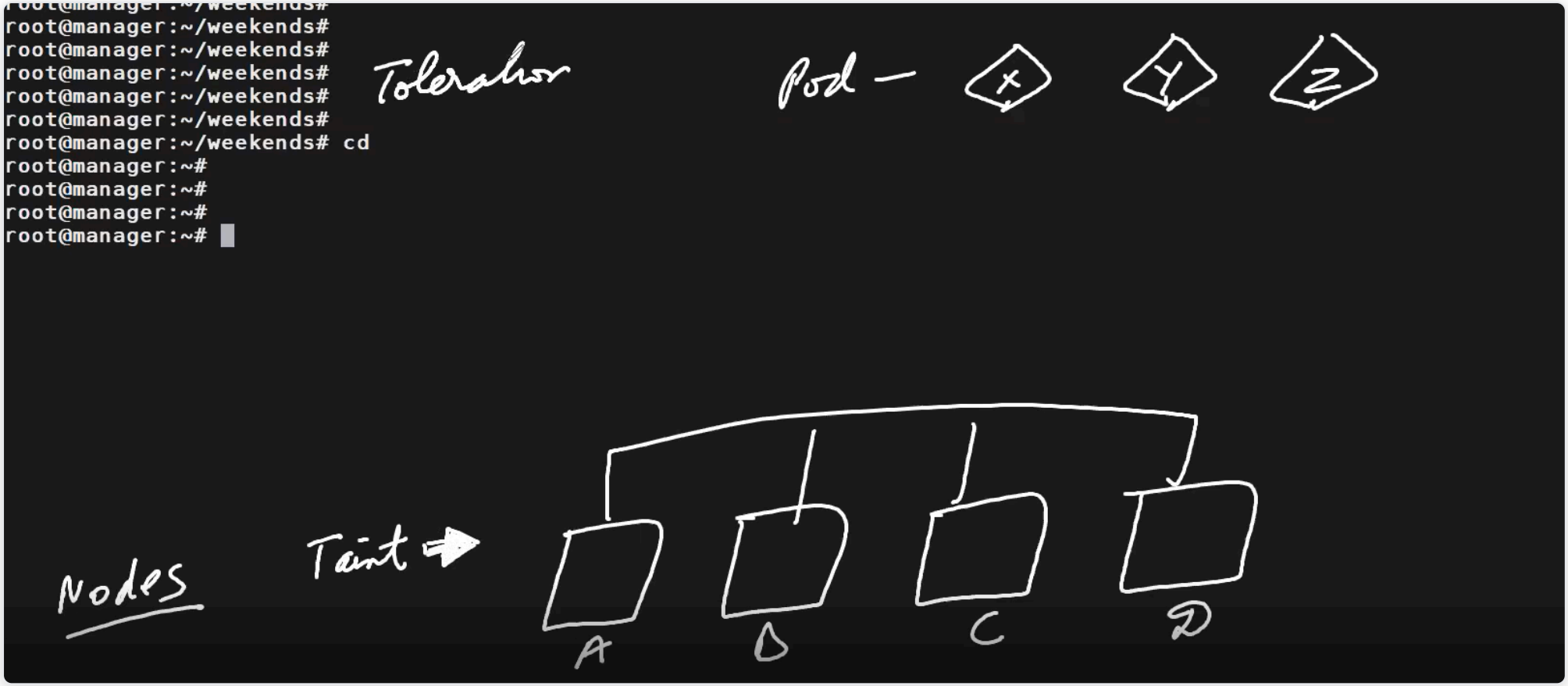
Raw screenshots captured while watching this session, kept alongside the written notes (07-taints-and-tolerations.md) for visual reference.

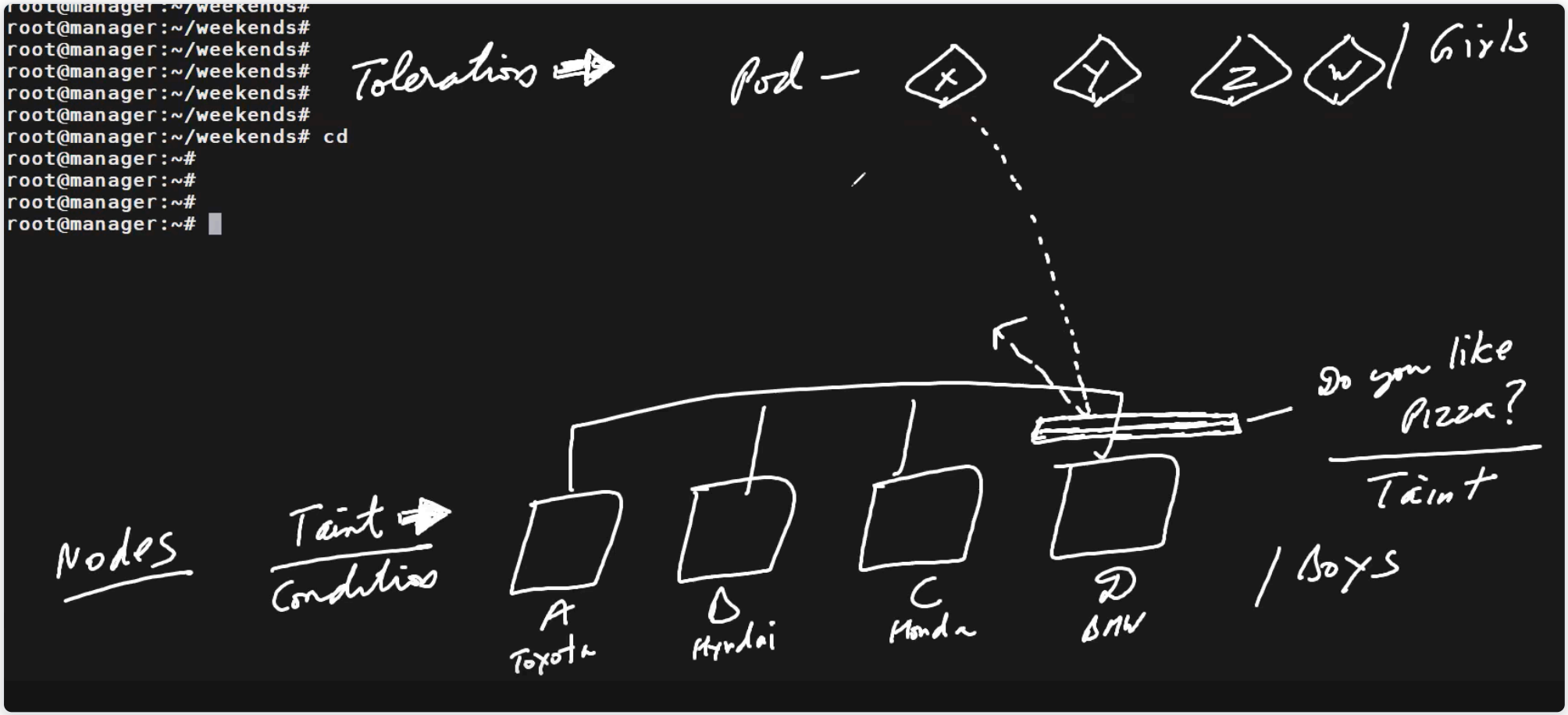
23 screenshots · captured 2026-09-16, 10:20 PM - 11:12 PM

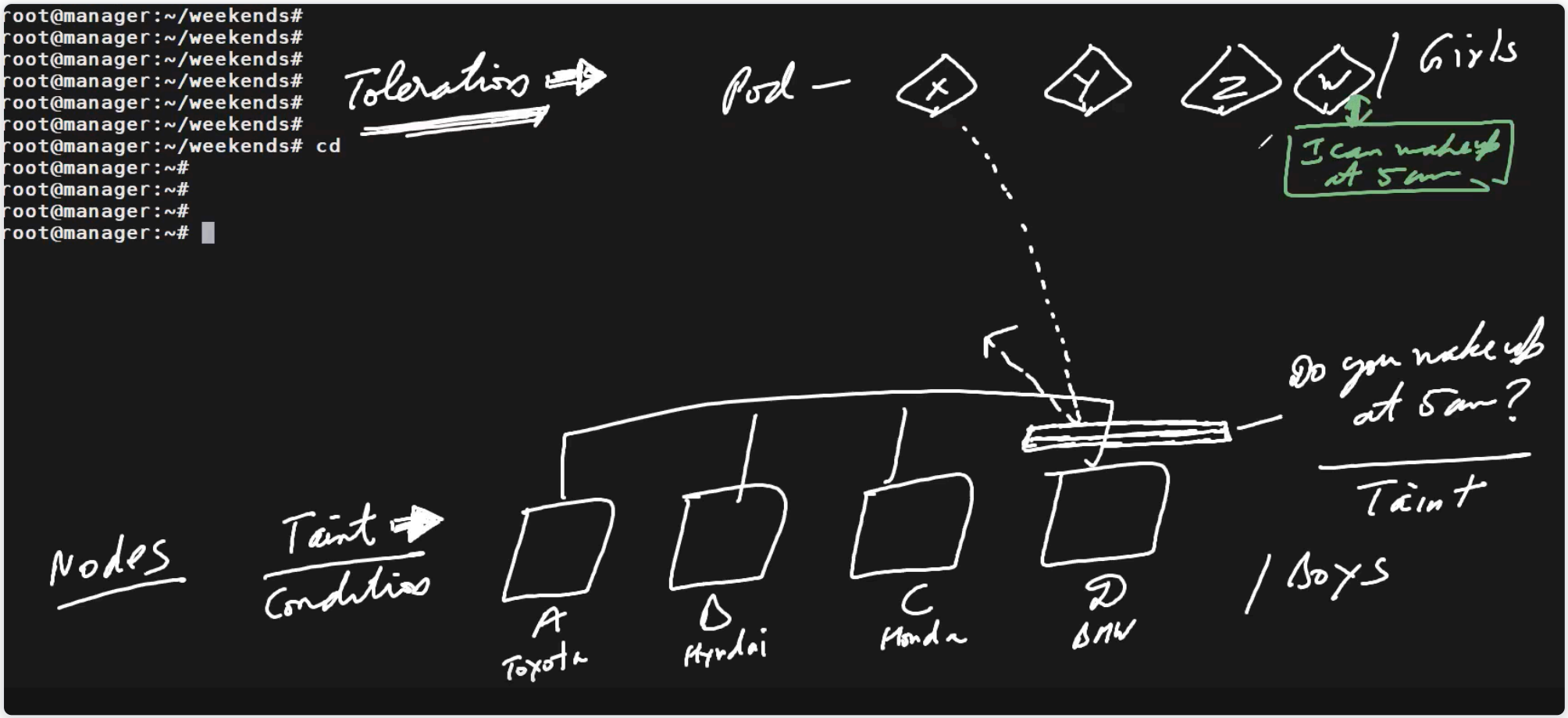

Taints & Tolerations
Node accepts only
certain kind of

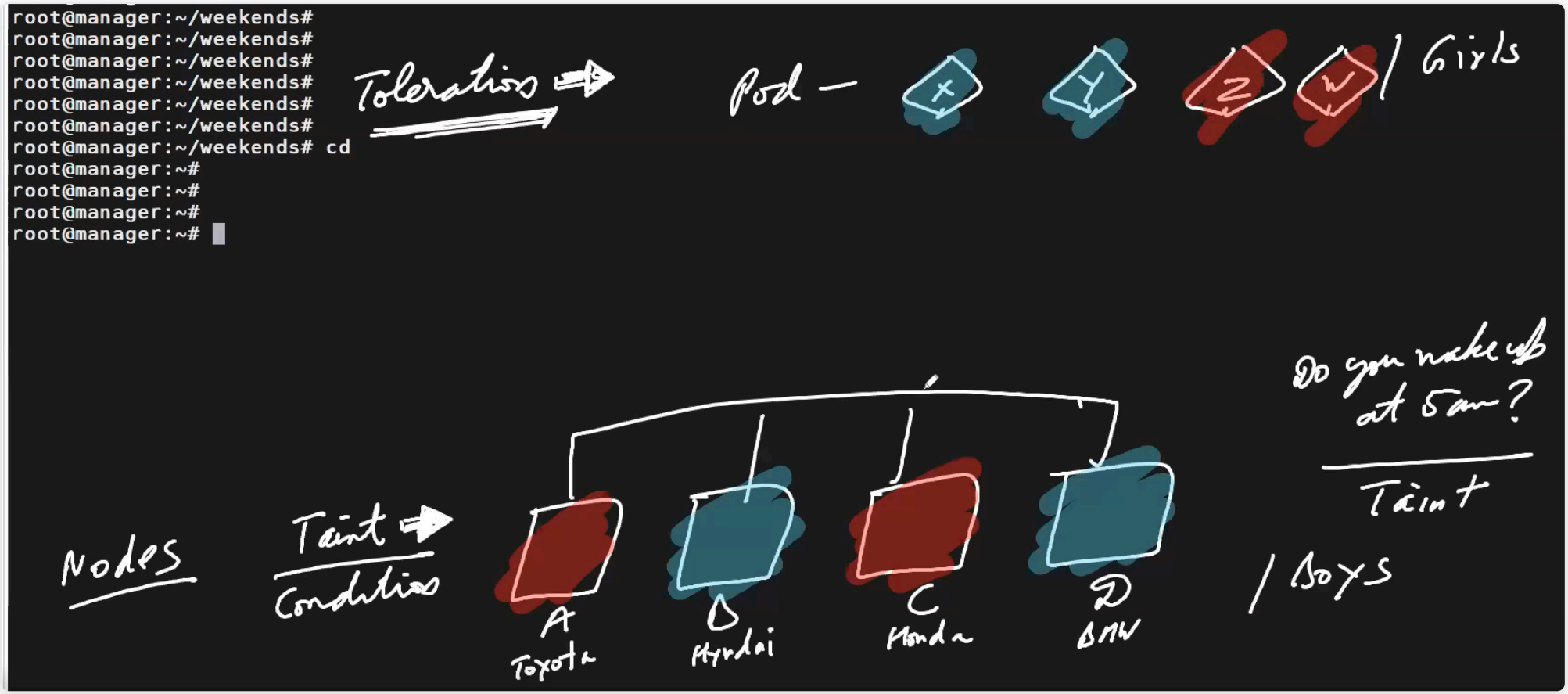

Taints & Tolerations
Node accepts only
certains kind of Pods

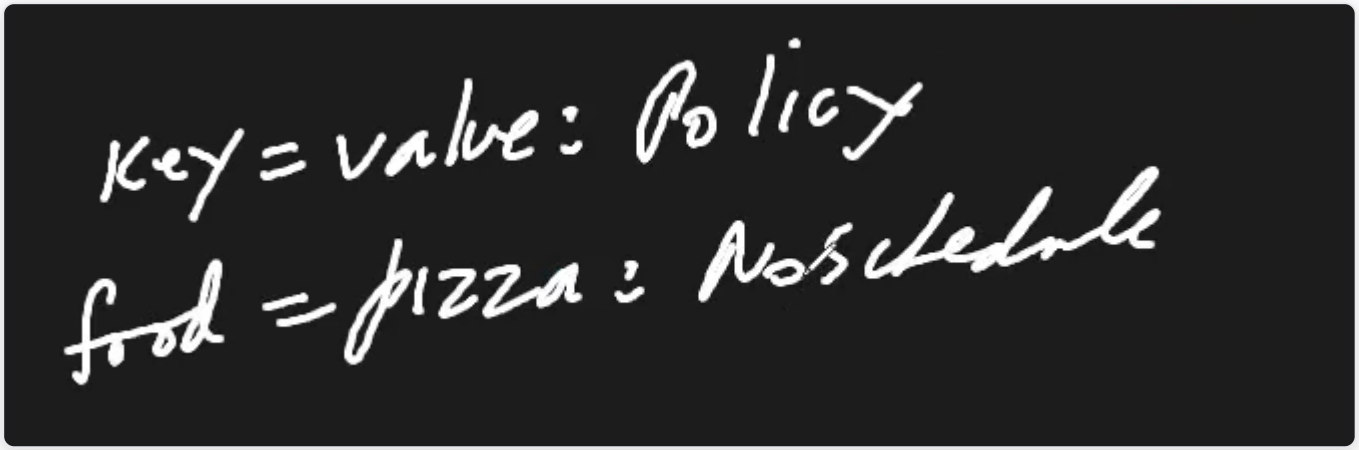

root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends# cd
root@manager:~#
root@manager:~#
root@manager:~#
root@manager:~#
Pod –
X
Y
Z
Nodes
A
B
C
D

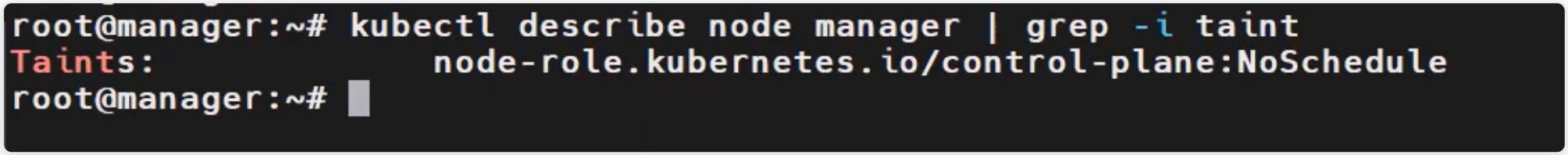

root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends# cd
root@manager:~#
root@manager:~#
root@manager:~#
root@manager:~#
Toleration
Pod –
X
Y
Z
Nodes
Taint
A
B
C
D


root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends# cd
root@manager:~#
root@manager:~#
root@manager:~#
root@manager:~#
Tolerations
Pod –
X
Y
Z
W
Girls
Do you like Pizza?
Taint
Nodes
Taint
Conditions
A
Toyota
B
Hyundai
C
Honda
D
BMW
Boys


root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends# cd
root@manager:~#
root@manager:~#
root@manager:~#
root@manager:~#
Tolerations
Pod –
X
Y
Z
W
Girls
I can make up at 5 am
Do you make up at 5 am?
Taint
Nodes
Taint
Conditions
A
Toyota
B
Hyundai
C
Honda
D
BMW
Boys

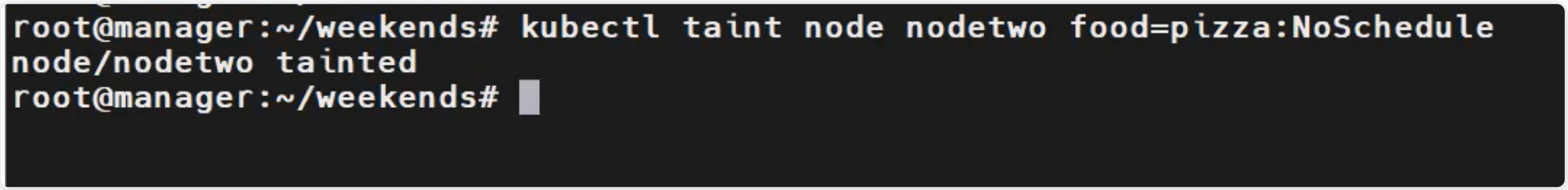

root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends# cd
root@manager:~#
root@manager:~#
root@manager:~#
root@manager:~#
Tolerations
Pod –
X
Y
Z
W
Girls
Do you wake up at 5 am?
Taint
Nodes
Taint
Conditions
A
Toyota
B
Hyundai
C
Honda
D
BMW
Boys


key = value : Policy
food = pizza : No schedule


root@manager:~# kubectl describe node manager | grep -i taint
Taints: node-role.kubernetes.io/control-plane:NoSchedule
root@manager:~#


root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends# cd
root@manager:~#
root@manager:~#
root@manager:~#
root@manager:~# kubectl describe node manager | grep -i taint
Taints: node-role.kubernetes.io/control-plane:NoSchedule
root@manager:~#
Key=value: Policy


kubectl taint node nodetwo food=pizza:NoS

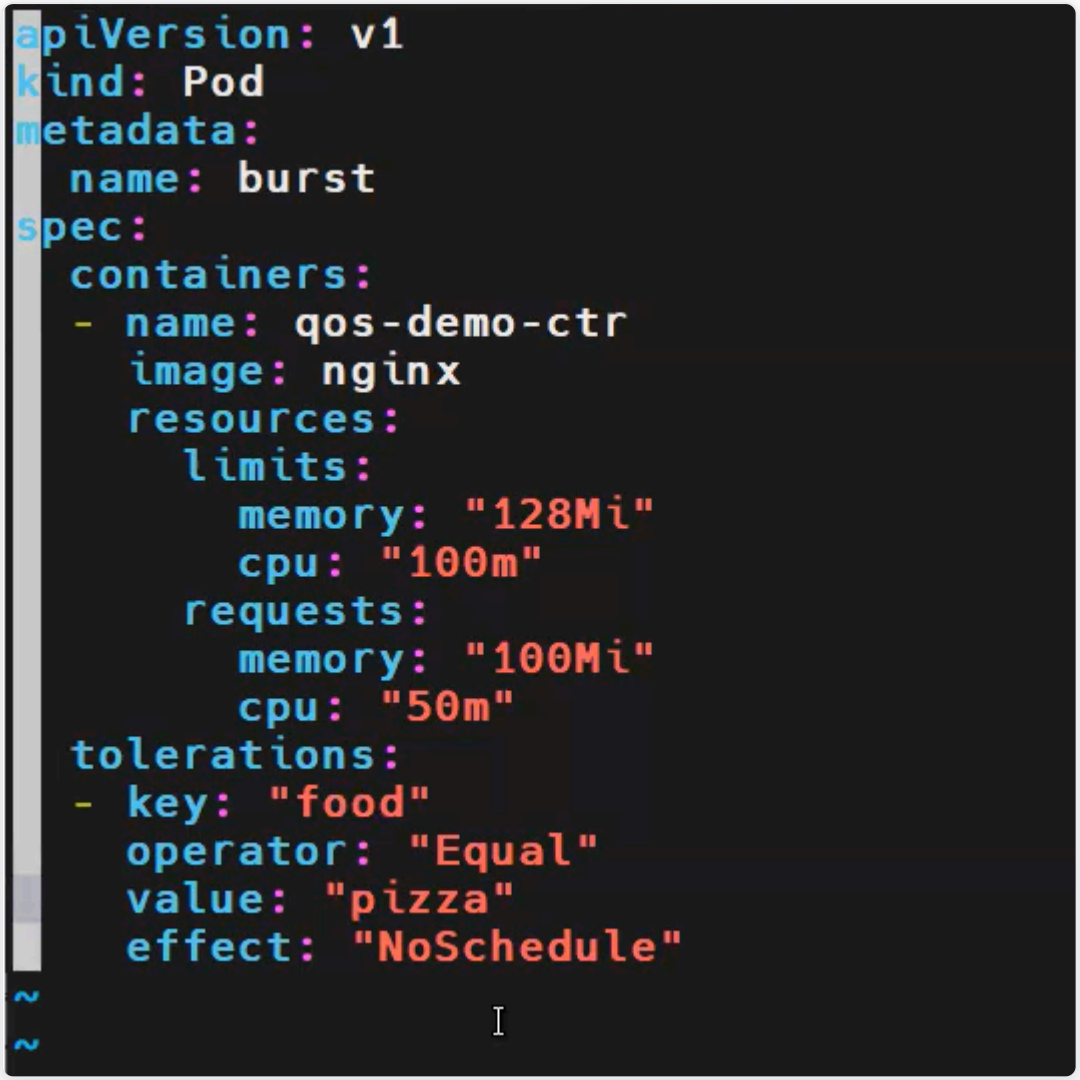

root@manager:~/weekends# kubectl taint node nodetwo food=pizza:NoSchedule
node/nodetwo tainted
root@manager:~/weekends#

```
root@manager:~/weekends# kubectl taint node nodetwo food=pizza:NoSchedule
node/nodetwo tainted
root@manager:~/weekends# kubectl describe node manager | grep -i taint
Taints:             node-role.kubernetes.io/control-plane:NoSchedule
root@manager:~/weekends# kubectl describe node nodetwo | grep -i taint
Taints:             food=pizza:NoSchedule
root@manager:~/weekends#
```

```yaml
tolerations:
- key: "key1"
  operator: "Equal"
  value: "value1"
  effect: "NoSchedule"
```

nodename	name of the node
key=value	some key value pair
taint-effect	NoSchedule or PreferNoSchedule or NoExecute

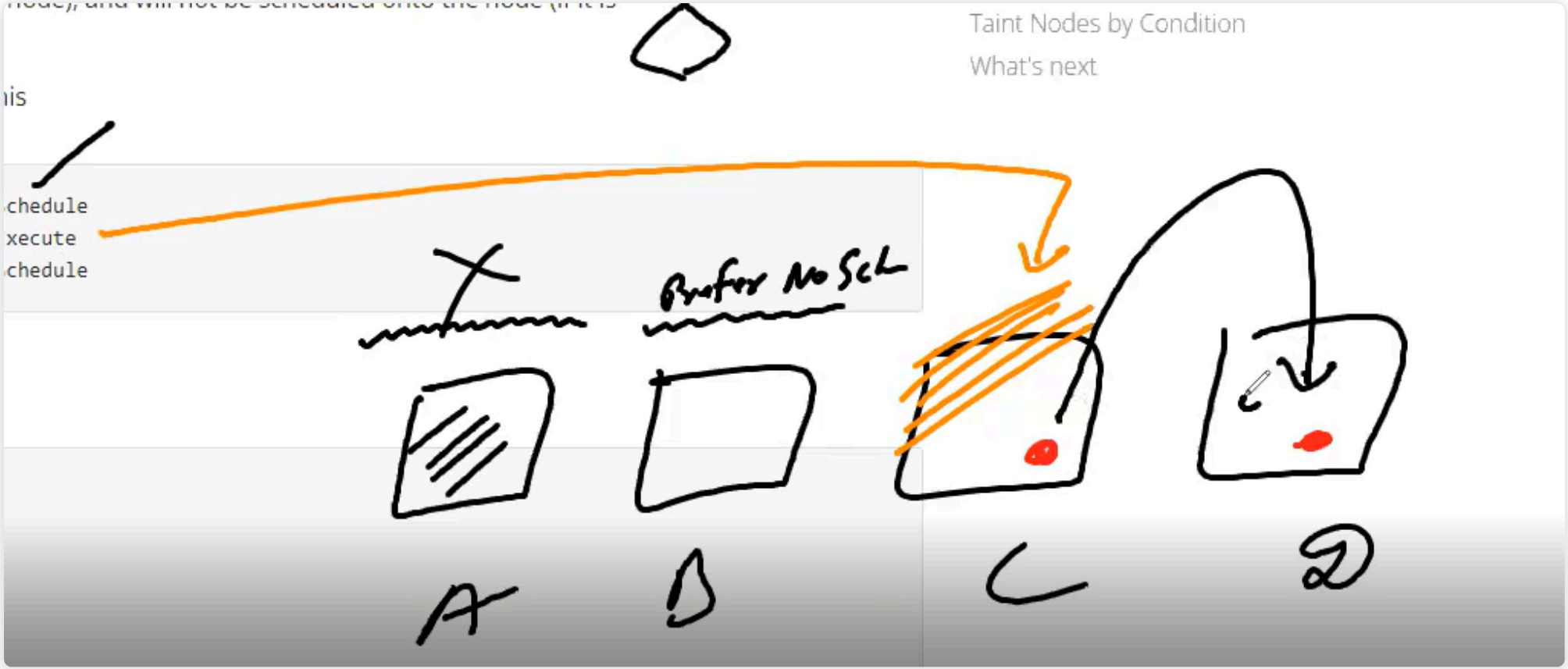
```yaml
apiVersion: v1
kind: Pod
metadata:
        name: myapp1
        labels:
                app: myapp
spec:
        containers:
                - name: ngnix-container
                  image: nginx
#        nodeName: node2.example.com
        tolerations:
        - key: "color"
          operator: "Equal"
          value: "pink"
          effect: "NoSchedule"
```

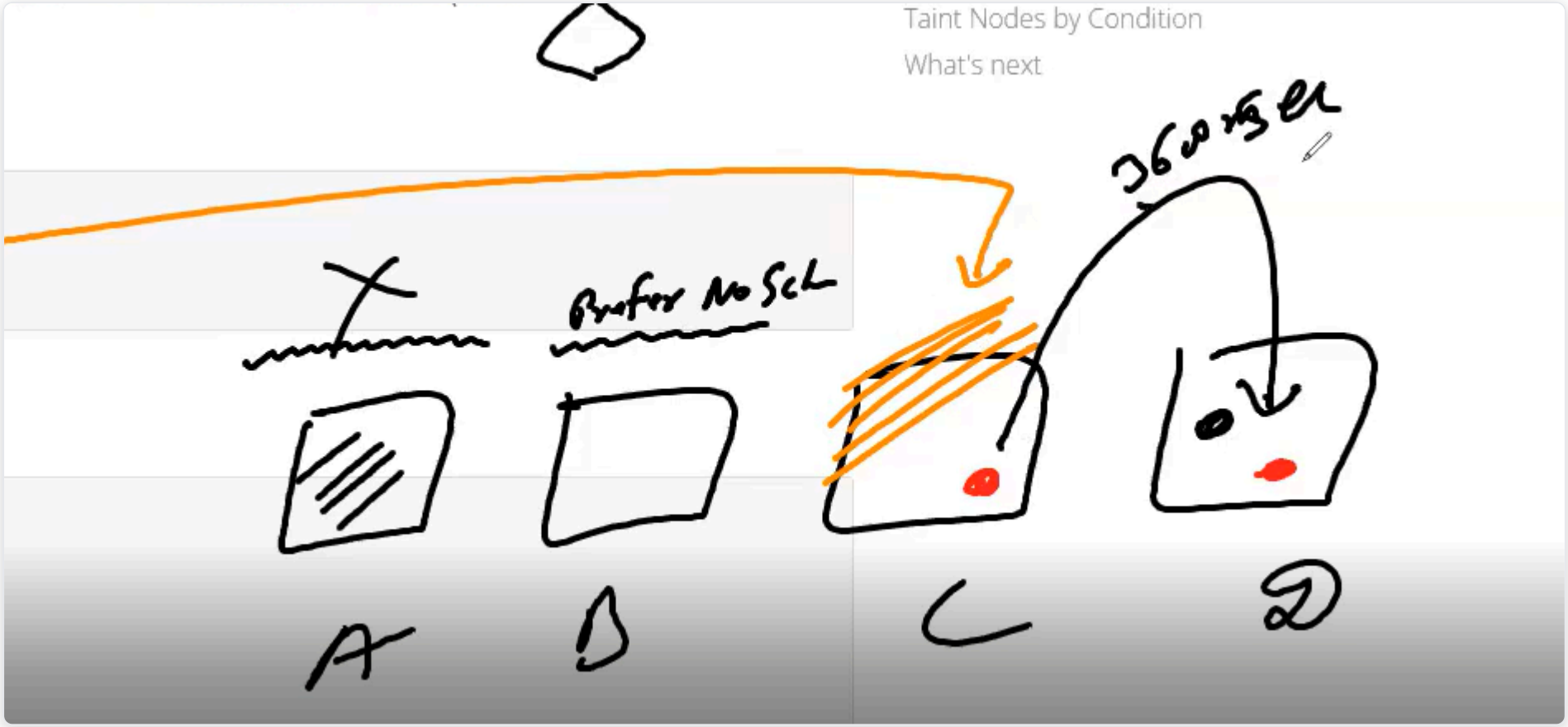

apiVersion: v1
kind: Pod
metadata:
name: burst
spec:
containers:
- name: qos-demo-ctr
image: nginx
resources:
limits:
memory: "128Mi"
cpu: "100m"
requests:
memory: "100Mi"
cpu: "50m"
tolerations:
- key: "food"
operator: "Equal"
value: "pizza"
effect: "NoSchedule"

```
    ConfigMapName:           kube-root-ca.crt
    ConfigMapOptional:       <nil>
    DownwardAPI:             true
QoS Class:                   Burstable
Node-Selectors:              <none>
Tolerations:                 food=pizza:NoSchedule
                             node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type     Reason     Age    From               Message
```

For example, imagine you taint a node like this

```
kubectl taint nodes node1 key1=value1:NoSchedule
kubectl taint nodes node1 key1=value1:NoExecute
kubectl taint nodes node1 key2=value2:NoSchedule
```


Taint Nodes by Condition
What's next
chedule
xecute
chedule
A
D
C
2


Taint Nodes by Condition
What's next
A
B
C
D

```yaml
tolerations:
- key: "key1"
  operator: "Equal"
  value: "value1"
  effect: "NoExecute"
  tolerationSeconds: 3600
```

means that if this pod is running and a matching taint is added to the node, then the pod will stay bound to the node for 3600 seconds, and then be evicted. If the taint is removed before that time, the pod will not be evicted.